Kishal Tandel

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Undergraduate Physics Student | Theoretical Physics, Quantum Dynamics, and Quantum Information

Education

August, 2023 – present **BS-MS in Physical Sciences**, Indian Institute of Science Education and Research, Kolkata, India

CGPA : 8.34/10 (after 5th Semester)

2023 **ISC**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India

Class 12 - 93.25%, Indian School Certificate (ISC)

2021 **ICSE**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India

Class 10 - 95.40%, Indian Certificate of Secondary Examination (ICSE)

Research Interests

Quantum Mechanics, Quantum Information, Theoretical and Computational Physics, Quantum Many-Body Systems, and Numerical Modelling of Quantum Dynamics.

I am particularly interested in analytical and numerical approaches to wave phenomena, quantum systems and quantum encryption.

Independent Research

April, 2026 **Wigner Quasiprobability Distribution**

Developed a detailed phase-space formulation of quantum mechanics using Weyl quantization and the Wigner formalism.

Explored connections between operator ordering, quasiprobability distributions, and semiclassical limits.

Derived the Wigner function from density matrix methods and studied its normalization, marginal distributions, invertibility, and time evolution equations.

Analyzed harmonic oscillator states, quantum interference effects, and quantum tomography via the Radon transform.

Derived time evolution equation of the Wigner function and visualized it for a superposition state of the harmonic oscillator.

April, 2025 **One-Dimensional Quantum Tunneling**

Solved the one-dimensional time-independent Schrödinger equation for finite potential barriers across all energy regimes.

Derived analytical expressions for transmission and reflection coefficients and proved continuity at the threshold energy $E = V_0$.

Implemented numerical visualization of tunneling behavior using Python and compared quantum mechanical predictions with classical transport behavior.

Projects

May, 2025 **Numerical Simulation of Three-Body Gravitational Dynamics**

Developed a fourth-order Runge–Kutta simulation framework in C++ for planar three-body gravitational systems.

Investigated sensitivity to initial conditions, orbital instability, and chaotic trajectories in nonlinear dynamical systems.

Generated numerical visualizations of long-term orbital evolution and emergent chaotic behavior.

Skills

Programming Python (Numpy, Scipy, Matplotlib), C++, C, Java

Language English, Hindi, Gujarati

Homepage kishaltandel.github.io